

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	T Deekshith	Reg. No.:	192472253
Section / Batch:	2024	Date:	01-08-2026

Code of Conduct

I T Deekshith (192472253) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deekshith

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

Q1. Morphological Analysis for Document Indexing

Input Words: connected, connecting, connection

Pipeline:

1. Tokenize input.
2. Identify root and suffix.
3. Apply rule-based decomposition.
4. Classify as inflectional or derivational.
5. Normalize to a common base form for indexing.

Word	Root	Suffix	Rule Applied	Type	Normalized
connected	connect	-ed	Remove -ed	Inflectional	connect
connecting	connect	-ing	Remove -ing	Inflectional	connect
connection	connect	-ion	Remove -ion	Derivational	connect

Parsed Structure:

connected = connect + ed

connecting = connect + ing

connection = connect + ion

Result: All words are normalized to the base form 'connect', enabling efficient document indexing.

Q2. Morphological Parsing for Sentiment Analysis

Input Words: unhappy, happiness, happily

Word	Prefix	Root	Suffix	Transformation	Meaning
unhappy	un-	happy	-	Derivational	Negation
happiness	-	happy	-ness	Derivational	State/quality
happily	-	happy	-ly	Derivational	Adverb

Semantic consistency: All forms share the same lexical base 'happy'. This helps sentiment analysis treat them as related words.

Q3. Stemming-Based Preprocessing

Input Words: played, player, playing

Word	Rule	Intermediate	Stem	Classification
played	Remove -ed	play+ed	play	Inflectional
player	Remove -er	play+er	play	Derivational
playing	Remove -ing	play+ing	play	Inflectional

Final stem extracted for all words: play. This ensures all grammatical variants retrieve the same documents.

Q4. Finite-State Morphological Parsing

Input Words: writes, writing, written

Word	Morphological Breakdown	Pattern	State	Normalized Root
writes	write+s	Regular	q0→q1→q2	write
writing	write+ing	Regular	q0→q1→q3	write
written	write+en	Irregular	q0→q1→q4	write

Finite-state parser recognizes both regular and irregular verb forms and maps them to the root 'write'.

Q5. Porter Stemmer-Based Processing

Input Words: relational, relation, relate

Word	Step 1	Step 2	Final Stem
relational	relate	remove final e	relat
relation	relate	remove final e	relat
relate	remove final e	-	relat

Porter stemming removes derivational suffixes step by step while preserving the common stem. Final stem: relat.

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
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Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____